

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Step 3 – Identify the Appropriate Regulatory Pathway

The major pathways students should know are:

A. 510(k) – Premarket Notification

The manufacturer generally demonstrates that the new device is **substantially equivalent** to a legally marketed predicate device.

The comparison generally considers:

- Intended use
- Technological characteristics
- Performance
- Safety

FDA explains that a 510(k) is a pathway based on substantial equivalence to an appropriate legally marketed device.

Important: The correct term is generally "**FDA clearance**", not "FDA approval," for a 510(k).

B. De Novo Classification

The **De Novo pathway** can be used for certain novel devices for which there is no suitable predicate but whose risks can be appropriately controlled.

A successful De Novo request establishes a new classification for that device type. FDA identifies De Novo as a pathway for certain novel device types that have not previously been marketed in the United States.

C. Premarket Approval – PMA

PMA is the most stringent major premarket pathway.

It is generally associated with **Class III devices** requiring PMA.

The manufacturer must provide scientific evidence supporting the safety and effectiveness of the device. FDA describes PMA as the scientific and regulatory review used to evaluate the safety and effectiveness of applicable Class III devices.

Examples of devices that may require PMA include certain implantable and life-supporting devices.

D. HDE

Humanitarian Device Exemption (HDE) is associated with certain devices intended to benefit patients with rare diseases or conditions meeting the applicable statutory criteria.

Step 4 – Preclinical/Nonclinical Testing

Before clinical use or marketing, applicable devices may require testing such as:

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Mechanical testing
- Electrical safety testing
- Biocompatibility
- Software verification
- Electromagnetic compatibility
- Sterilization validation
- Packaging testing
- Performance testing
- Animal testing where appropriate

The exact testing depends on the device.

Step 5 – Clinical Evaluation/Investigation

For devices requiring clinical evidence, clinical investigations may be conducted to evaluate:

- Safety
- Effectiveness
- Performance
- Patient outcomes
- Adverse events

Where applicable, an **Investigational Device Exemption (IDE)** permits certain investigational devices to be used in clinical studies in accordance with FDA requirements.

Step 6 – Quality Management System

Manufacturers must establish an appropriate quality system.

A major current development is that FDA's **Quality Management System Regulation (QMSR)** became effective on **February 2, 2026**. QMSR amended 21 CFR Part 820 and incorporates by reference **ISO 13485:2016** for medical device quality management systems.

Therefore, students should now understand:

21 CFR Part 820 → QMSR → incorporates ISO 13485:2016

The QMS supports areas such as:

- Design and development
- Manufacturing
- Quality control
- Documentation

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Corrective and preventive actions
- Complaint handling
- Risk management
- Records

Step 7 – Prepare Regulatory Submission

The manufacturer prepares the appropriate regulatory submission.

Depending on the pathway, this may include:

- Device description
- Intended use
- Indications for use
- Design information
- Risk information
- Performance data
- Test results
- Clinical data where required
- Manufacturing information
- Quality-system information
- Labeling

Step 8 – Labeling

Medical device labeling is an important regulatory requirement.

FDA's general device labeling requirements are contained in **21 CFR Part 801**.

Labeling may include:

- Device name
- Manufacturer information
- Intended use
- Indications
- Instructions for use
- Warnings
- Precautions
- Contraindications

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Installation instructions
- Maintenance information

Labeling must not provide misleading information or unsupported claims.

Step 9 – FDA Review

FDA reviews the submitted information according to the applicable pathway.

The review may involve:

- Technical evaluation
- Clinical evaluation
- Statistical evaluation
- Manufacturing/quality considerations
- Labeling review
- Risk assessment

FDA may request additional information where necessary.

Step 10 – Regulatory Decision

Depending on the pathway, the outcome may be:

- 510(k) clearance
- De Novo classification
- PMA approval
- HDE approval
- Exemption where applicable

Step 11 – Post-Market Requirements

FDA oversight does not end when a device reaches the market.

Post-market activities may include:

- Complaint handling
- Medical device reporting
- Corrective actions
- Recalls where necessary
- Post-market surveillance
- Manufacturing quality monitoring

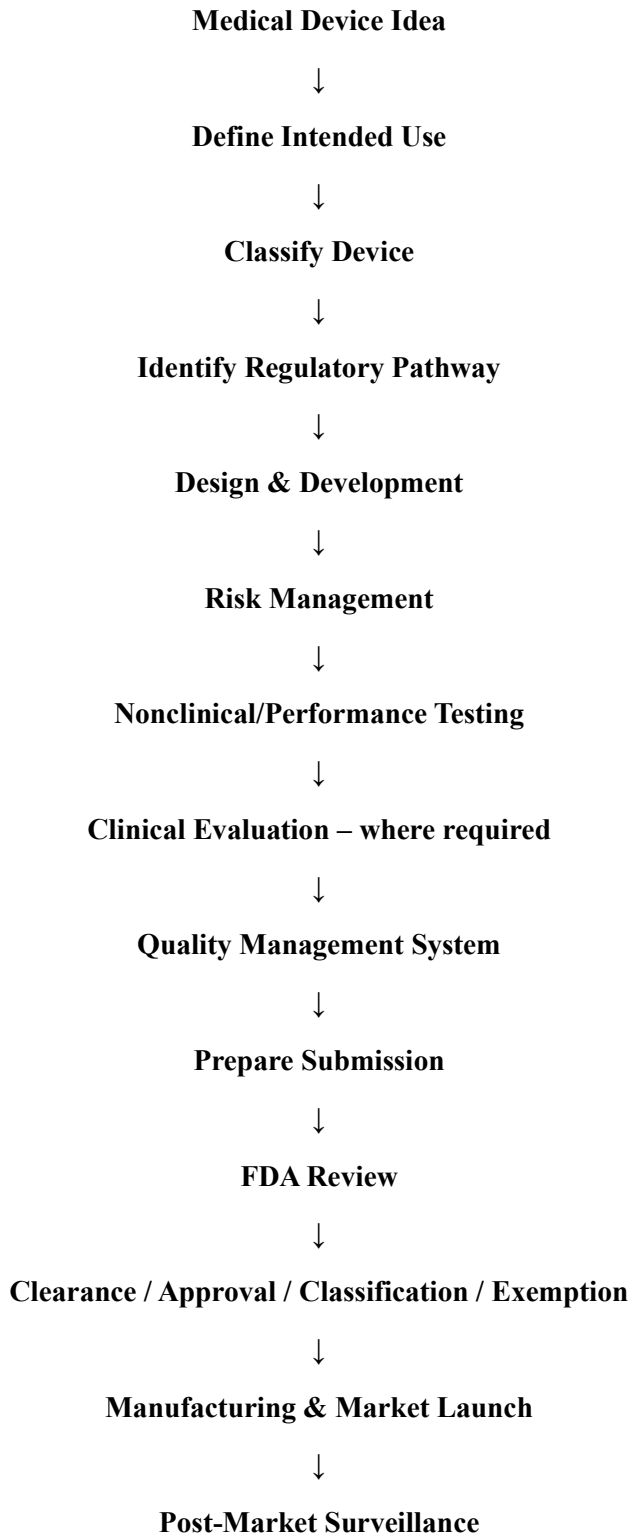
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UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Monitoring of device safety and performance

FDA states that it is responsible for assuring that medical devices available in the United States remain safe and effective throughout their total product lifecycle.

Simplified FDA Pathway



MEDICAL DEVICE MARKETING
UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Important Difference: Clearance vs Approval

Term	Common pathway	Meaning
FDA Clearance	510(k)	Device has been determined substantially equivalent to an appropriate predicate
FDA Approval	PMA	FDA has approved the PMA based on applicable evidence of safety and effectiveness
De Novo Classification	De Novo	New device type classified through the De Novo pathway
Exempt	Certain devices	No applicable premarket submission, subject to the conditions and limitations of the exemption

5. Reimbursement

Introduction

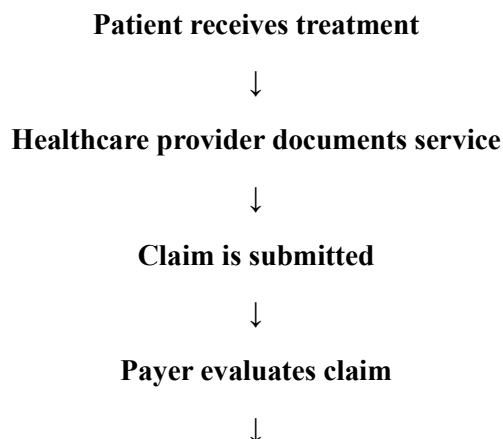
Reimbursement refers to the payment made to healthcare providers or organizations for eligible healthcare services, procedures, or products.

For a medical device company, reimbursement is important because a product may be clinically effective but commercially unsuccessful if customers cannot obtain adequate financial coverage for its use.

Major Participants

- Patient
- Doctor
- Hospital
- Insurance company/payer
- Government
- Medical device manufacturer

Basic Process



MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Claim approved/rejected



Payment made

Importance of Reimbursement

Reimbursement influences:

- Patient affordability
- Hospital purchasing
- Product adoption
- Market size
- Revenue
- Product pricing

Example

A new cardiac monitoring device may provide excellent clinical benefits. However, if hospitals or patients cannot recover the cost through an applicable payment mechanism, adoption may remain low.

6. Managing Issues

During product development and commercialization, several issues can arise.

Common issues

- Technical failures
- Component shortages
- Design changes
- Budget overruns
- Regulatory delays
- Testing failures
- Customer complaints
- Manufacturing problems
- Software errors
- Supplier delays

Issue Management Process

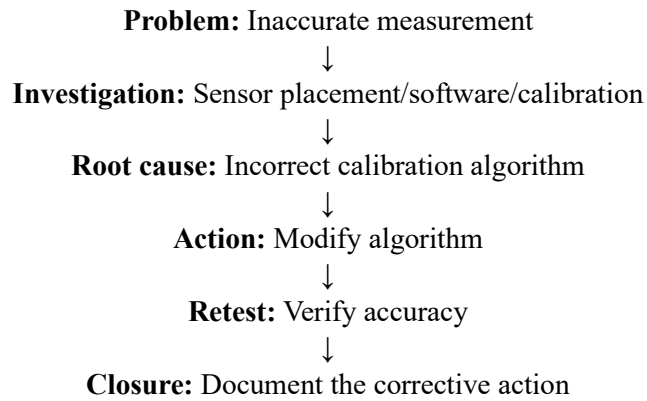
Identify issue → Analyze cause → Determine impact → Prioritize → Develop corrective action → Assign responsibility → Implement action → Monitor → Close issue

Example

If a prototype shows inaccurate SpO₂ readings:

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UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH



7. Meeting Deadlines

Meeting deadlines means completing project activities within the planned time.

Methods

1. Develop a project schedule

Define:

- Activities
- Start date
- End date
- Dependencies
- Responsible person

2. Prioritize critical activities

Identify activities that directly affect the final delivery date.

3. Monitor progress

Use:

- Gantt charts
- Milestone tracking
- Weekly meetings
- Progress reports

4. Allocate resources properly

Ensure availability of:

- Personnel
- Equipment
- Materials

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Budget

5. Manage risks

Identify potential delays before they occur.

6. Take corrective action

If an activity is delayed:

- Reallocate resources
- Change sequence
- Add manpower
- Revise schedule
- Address technical problems

Example

If clinical testing is delayed by two weeks, the team should assess whether documentation, software development, or manufacturing activities can proceed in parallel without compromising quality.

8. Product Launch

Introduction

Product launch is the process of introducing a new product into the target market and making it available to customers.

For a medical device, product launch is more than simply selling the product. It involves:

- Regulatory readiness
- Manufacturing readiness
- Market preparation
- Pricing
- Distribution
- Sales training
- Customer support
- Product communication
- Post-market monitoring

Stages of Product Launch

1. Market Preparation

Before launching the product, the company should understand:

- Target customers

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UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Market size
- Competitors
- Customer requirements
- Pricing
- Distribution channels
- Market barriers

2. Regulatory Readiness

The company must ensure that applicable regulatory requirements have been satisfied before marketing the device in the relevant jurisdiction.

This includes appropriate:

- Regulatory status
- Labeling
- Documentation
- Quality systems
- Manufacturing controls

3. Manufacturing Readiness

The company should ensure that sufficient quantities can be produced with consistent quality.

Activities include:

- Supplier qualification
- Production planning
- Quality inspection
- Packaging
- Sterilization where required
- Inventory planning

4. Pricing Strategy

The company determines the price based on:

- Manufacturing cost
- Competitor prices
- Customer willingness to pay
- Clinical value

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Distribution cost
- Market positioning
- Reimbursement environment

5. Sales Preparation

The sales team should understand:

- Product features
- Clinical benefits
- Product limitations
- Target customers
- Competitors
- Pricing
- Demonstration procedure
- Customer objections

6. Marketing and Promotion

Promotion may include:

- Product demonstrations
- Conferences
- Medical exhibitions
- Digital marketing
- Scientific publications
- Healthcare professional education
- Sales presentations

Medical device claims should remain consistent with applicable regulatory requirements and authorized/intended uses.

7. Distribution

The company determines how the product will reach customers.

Possible channels include:

Manufacturer → Distributor → Hospital → Patient

or

Manufacturer → Hospital

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UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

or

Manufacturer → Clinic → Patient

8. Product Training

Healthcare professionals may need training on:

- Installation
- Operation
- Maintenance
- Safety
- Troubleshooting
- Clinical use

9. Launch

The product is officially introduced to the target market.

Launch activities may include:

- Product announcement
- Demonstrations
- Distributor activation
- Sales campaigns
- Hospital visits
- Customer training

10. Post-Launch Monitoring

After launch, the company monitors:

- Sales
- Customer feedback
- Complaints
- Device failures
- Returns
- Adverse events
- Market acceptance
- Profitability

MEDICAL DEVICE MARKETING UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Product Launch Framework



Factors Determining Successful Product Launch

Technical readiness

The device should perform according to its specifications.

Clinical readiness

Healthcare professionals should understand its clinical value.

Regulatory readiness

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Required regulatory requirements must be satisfied.

Financial readiness

The company should have adequate financial resources.

Manufacturing readiness

Production should meet expected demand.

Sales readiness

Sales personnel should be properly trained.

Customer support

Installation, training, maintenance, and troubleshooting should be available.

Market readiness

There should be sufficient customer demand.

9. Forecasting

Forecasting is the process of **estimating future demand, sales, revenue, or market requirements**.

It helps companies determine how many products they should manufacture and how much inventory they should maintain.

Types of Forecasting

1. Qualitative forecasting

Based on expert opinion and market knowledge.

Methods include:

- Expert opinion
- Sales-force opinion
- Customer surveys
- Delphi method

2. Quantitative forecasting

Uses historical data and mathematical/statistical techniques.

Examples:

- Moving averages
- Trend analysis
- Regression analysis
- Time-series analysis

Factors Affecting Medical Device Forecasting

- Number of hospitals

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Patient population
- Disease prevalence
- Competitor products
- Product price
- Healthcare policies
- Reimbursement
- Technology adoption
- Economic conditions

Example

If a company sold:

- 500 units in Year 1
- 700 units in Year 2
- 900 units in Year 3

The company may use historical trends and market growth assumptions to estimate future demand.

Importance

Forecasting helps with:

- Production planning
- Inventory control
- Manpower planning
- Financial planning
- Sales targets
- Supply-chain management

10. Pricing

Pricing is the process of determining the **amount a customer must pay for a product or service.**

Objectives of Pricing

- Recover costs
- Generate profit
- Gain market share
- Compete effectively
- Reflect product value
- Encourage adoption

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Factors Affecting Pricing

1. Manufacturing cost

Higher production costs generally increase the minimum viable selling price.

2. Research and development cost

Medical device companies must recover development investment over time.

3. Competitor pricing

The company should understand prices of competing products.

4. Customer willingness to pay

Customers may pay more for significant clinical and economic benefits.

5. Clinical value

A product that significantly improves outcomes may justify a higher price.

6. Reimbursement

The available payment mechanism can strongly influence practical affordability and adoption.

7. Distribution cost

Distributor margins, transportation, installation, and service affect the final price.

Common Pricing Approaches

Cost-plus pricing:

Cost + desired profit margin

Competition-based pricing:

Price based on competitors

Value-based pricing:

Price based on perceived/ demonstrated customer and clinical value

Penetration pricing:

Lower initial price to encourage market adoption

Premium pricing:

Higher price for differentiated products with strong value or positioning

11. Product and Sales Support

Introduction

Product and sales support refers to the technical, clinical, marketing, training, and service assistance provided to customers and sales teams before and after a product is sold.

Product Support

Product support includes:

- Installation

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UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- User training
- Operation manuals
- Technical assistance
- Preventive maintenance
- Corrective maintenance
- Spare parts
- Software updates
- Troubleshooting
- Warranty service

Sales Support

Sales support helps sales personnel successfully communicate and demonstrate the product.

It includes:

- Product brochures
- Technical specifications
- Demonstration units
- Product presentations
- Training for sales personnel
- Competitor comparison
- Frequently asked questions
- Clinical evidence
- Customer case studies
- Pricing information

Importance

1. Improves customer confidence

Customers feel more confident when technical support is available.

2. Increases sales effectiveness

Sales personnel can answer technical and clinical questions.

3. Reduces product misuse

Proper training helps users operate devices correctly.

4. Improves customer satisfaction

Fast service and troubleshooting improve the customer experience.

5. Encourages repeat purchases

MEDICAL DEVICE MARKETING

UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

Satisfied customers are more likely to purchase additional products.

Example

For an ultrasound system, sales support may include product demonstration and clinical application training, while product support includes installation, preventive maintenance, troubleshooting, and service.

12. Managing an Existing Product Line

Introduction

A **product line** is a group of related products marketed by the same company.

For example, a medical technology company may have a product line containing:

- Basic patient monitor
- Advanced patient monitor
- ICU monitor
- Wireless monitor

Managing an existing product line means continuously monitoring and improving the products to maintain their **market position, profitability, quality, and customer satisfaction**.

Major Activities

1. Monitor sales performance

Track:

- Sales volume
- Revenue
- Profit
- Market share
- Growth rate

2. Monitor customer feedback

Collect information about:

- Product satisfaction
- Complaints
- Usability
- Reliability
- Desired improvements

3. Product improvement

Improve:

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UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Hardware
- Software
- User interface
- Accuracy
- Connectivity
- Reliability

4. Manage product variants

Companies may offer different versions for different customer segments.

5. Manage pricing

Prices may need adjustment based on:

- Competition
- Cost
- Demand
- Market position

6. Inventory management

Maintain sufficient stock without creating excessive inventory.

7. Competitor monitoring

Track:

- New competitor products
- Technology changes
- Pricing
- Features
- Market strategy

8. Product life-cycle management

A product generally passes through:

Introduction → Growth → Maturity → Decline

At each stage, the company may modify its marketing and product strategy.

9. Decide whether to modify or discontinue

If a product has declining demand, high maintenance costs, or is replaced by better technology, the company may:

- Improve it
- Reposition it
- Reduce its price

MEDICAL DEVICE MARKETING
UNIT – II: PRODUCT DEVELOPMENT AND LAUNCH

- Replace it
- Discontinue it